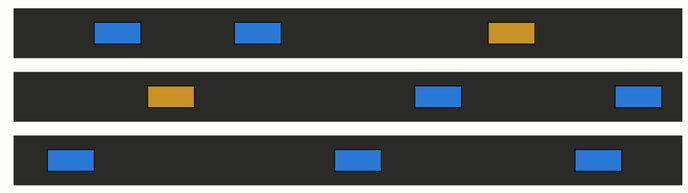
SELF-SOVEREIGN IDENTITY FOR CONNECTED VEHICLES



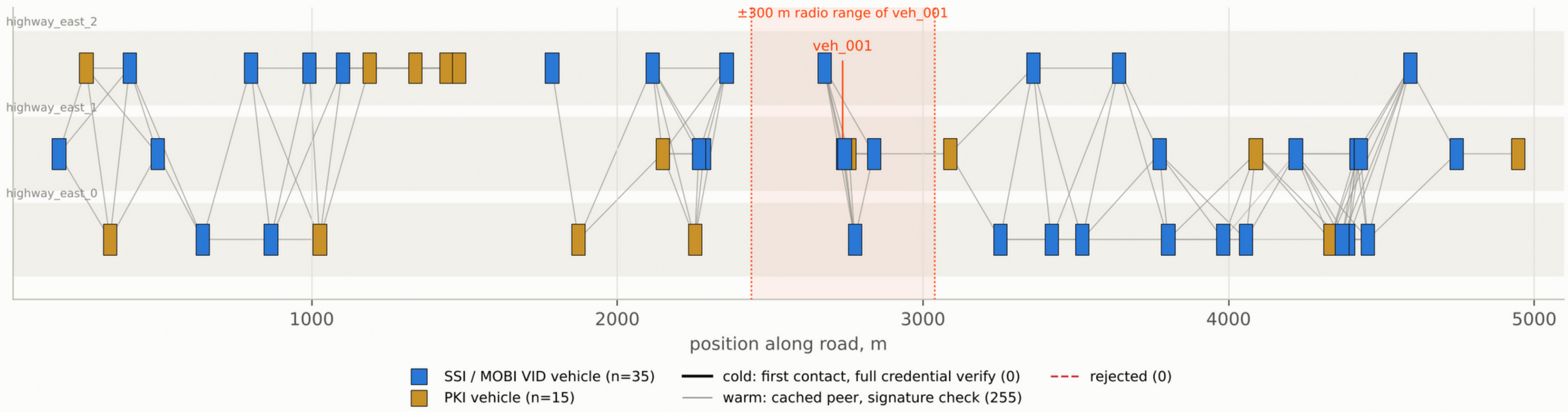
blue = SSI / MOBI VID (W3C credential, secp256k1)
ochre = PKI (X.509, ECDSA P-256)

Every V2V message signed. Every receiver verifying. The identity layer, shown.

Nikhil Prakash · MASc, Electrical and Computer Engineering, University of British Columbia

THE ROAD AT T = 15.0 S

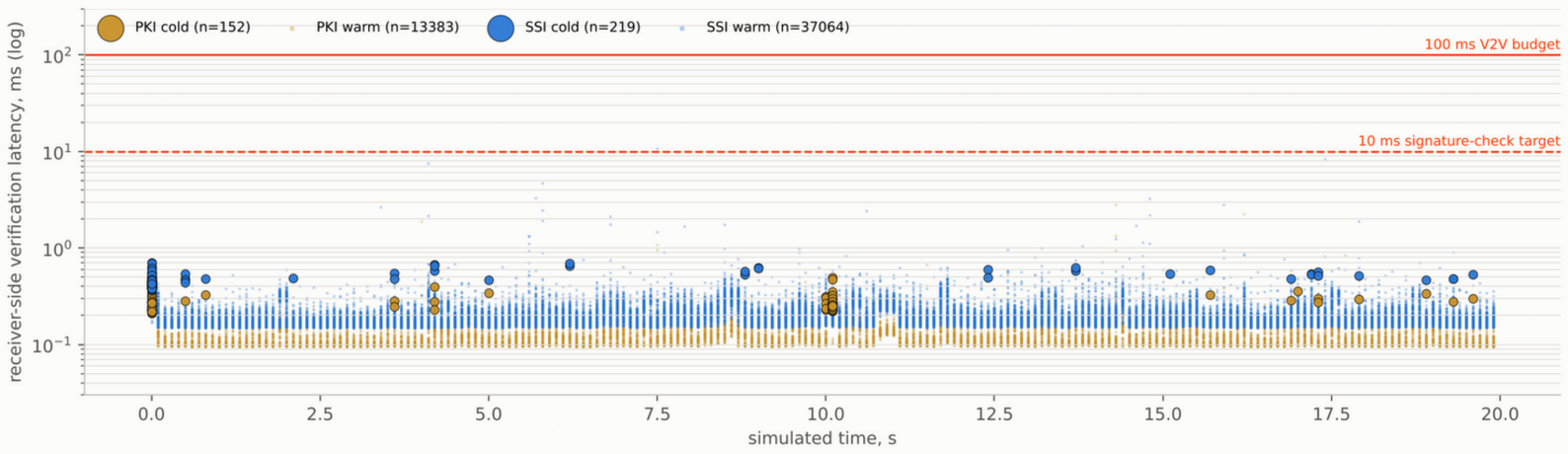
One 100 ms step: 51 signed BSM broadcasts, 255 receiver-side verifications, every one a real cryptographic operation.



trace sim_50v_20s_seed6.jsonl.gz · mode simulate · seed 6 · 20 s · 10 Hz BSM · 300 m radius · ≤8 rx/broadcast · commit 61d7157 · no radio channel, no MAC

EVERY VERIFICATION, AGAINST THE BUDGET

Each mark is one measured receiver-side verification (time.perf_counter around the cryptographic call). Compute latency only: no radio, MAC or channel loss is modelled.



trace sim_50v_20s_seed6.jsonl.gz · mode simulate · seed 6 · 20 s · 10 Hz BSM · 300 m radius · ≤8 rx/broadcast · commit 61d7157 · no radio channel, no MAC

ATTACK BOARD

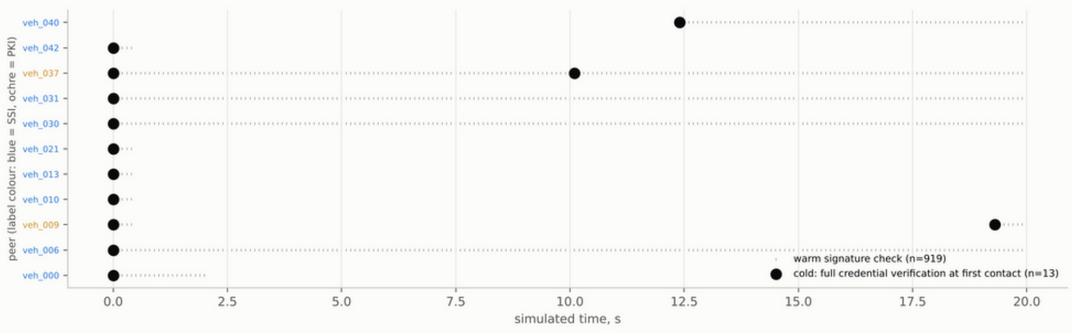
7/7 injected bad messages rejected at t = 19.9 s. Pass/fail is computed by the receiver's real verification path, not asserted.

Tampered PKI BSM speed field altered after signing	REJECTED
Tampered SSI BSM position altered after signing	REJECTED
Stale PKI BSM genuine signature, timestamp 5 s old	REJECTED
Stale SSI BSM genuine signature, timestamp 5 s old	REJECTED
Replayed PKI BSM accepted once, delivered again	REJECTED
Replayed SSI BSM accepted once, delivered again	REJECTED
Uncredentialed sender valid key, DID, no credential	REJECTED

trace sim_50v_20s_seed6.jsonl.gz · mode simulate · seed 6 · 20 s · 10 Hz BSM · 300 m radius · ≤8 rx/broadcast · commit 61d7157 · no radio channel, no MAC

VERIFY ONCE, CHECK EVERY MESSAGE

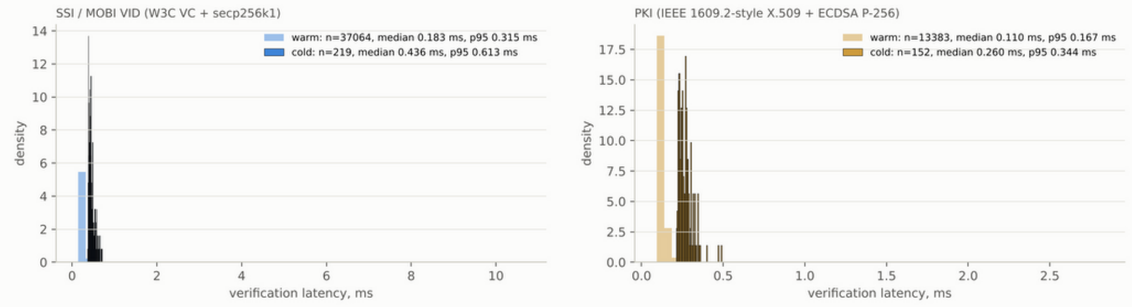
Receiver veh_045 (SSI / MOBI VID (W3C VC + secp256k1)). A peer's credential is fully verified at first contact; each later 10 Hz message is one signature check.
PKI peers re-enter cold whenever their pseudonym certificate rotates (every 100 messages); an SSI peer's DID binding is verified once.



trace sim_50v_20s_seed6.jsonl.gz · mode simulate · seed 6 · 20 s · 10 Hz BSM · 300 m radius · ≤8 rx/broadcast · commit 61d7157 · no radio channel, no MAC

COLD VS WARM

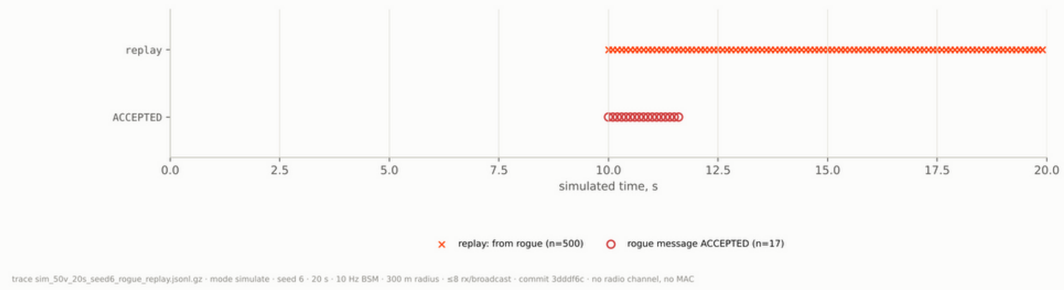
First-contact verification (cold) pays for the credential; every message after it (warm) pays for one signature check.



trace sim_50v_20s_seed6.jsonl.gz · mode simulate · seed 6 · 20 s · 10 Hz BSM · 300 m radius · ≤8 rx/broadcast · commit 61d7157 · no radio channel, no MAC

REJECTED, AND WHY

Every receiver-side rejection in the run, by the verifier's reason (500 in total). Rogue mode: replay; rogue messages accepted: 17. Attack-board probes after the run are not receiver traffic and are not drawn.



trace sim_50v_20s_seed6_rogue_replay.jsonl.gz · mode simulate · seed 6 · 20 s · 10 Hz BSM · 300 m radius · ≤8 rx/broadcast · commit 3dddf6c · no radio channel, no MAC

617 of 617 uncredentialed and 517 of 517 tampered rogue messages rejected.
17 replays accepted: fresh genuine messages relayed to receivers that missed the original. Authenticity holds; reach is extended.

WHAT THE PICTURES MAY CLAIM

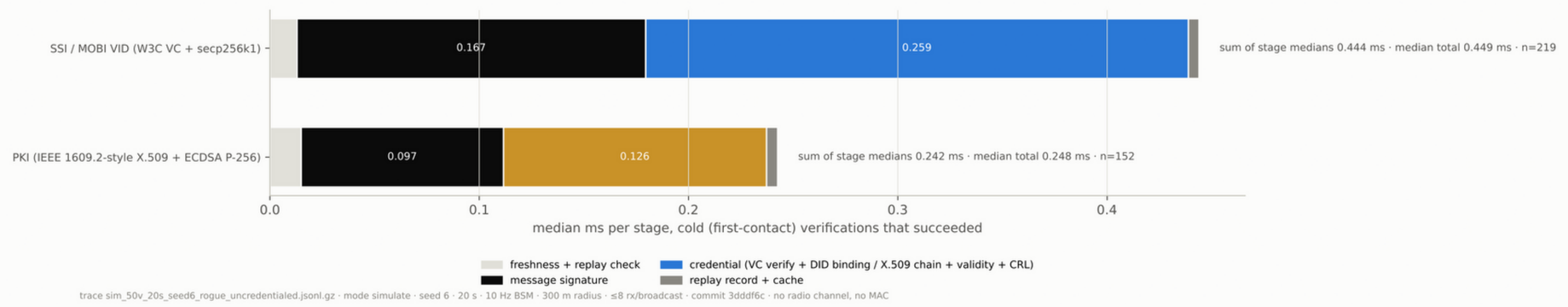
Signing, verification	real crypto, every message
Latency on any axis	real, perf_counter, compute only
Vehicle motion	mock model or SUMO 1.28 (TraCI)
Radio channel	mock: 8 nearest within 300 m
The blockchain	not in the V2V loop

0.142 ms

SSI warm verify, median of 30 run-medians (register #27), against a 100 ms V2V budget

WHERE COLD LATENCY GOES

Each first-contact verification split at its stage boundaries. The stage timers add their own perf_counter calls, so these totals sit slightly above the single-timer cold medians; they show proportions, not results of record (register #27).



trace sim_50v_20s_seed6_rogue_uncredentialed.jsonl.gz · mode simulate · seed 6 · 20 s · 10 Hz BSM · 300 m radius · ≤8 rx/broadcast · commit 3dddf6c · no radio channel, no MAC

Every figure regenerates from a committed trace: docs/visual/FIGURES.md · trace schema cvin-v2v-trace/1 · mock mobility unless marked SUMO · no radio channel or MAC modelled · interactive replay: docs/visual/replay/